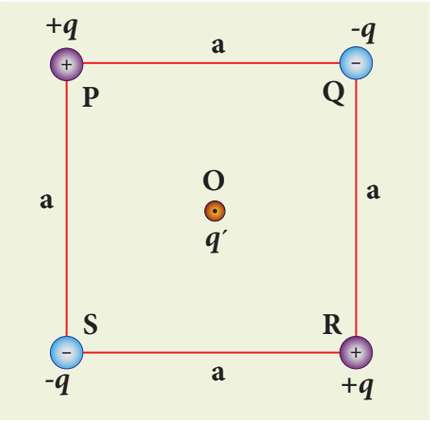
EXAMPLE 1.15

Four charges are arranged at the corners of the square *PQRS* of side *a* as shown in the figure.(a) Find the work required to assemble these charges in the given configuration. (b) Suppose a charge q′ is brought to the center of the square, by keeping the four charges fixed at the corners, how much extra work is required for this?



Solution

(a) The work done to arrange the charges in the corners of the square is independent of the way they are arranged. We can follow any order.

(i) First, the charge +q is brought to the corner P. This requires no work since no charge is already present, $W_P = 0$

(ii) Work required to bring the charge –q to the corner Q = (-q) × potential at a point Q due to +q located at a point P.

$$W_Q = -q \times \frac{1}{4\pi\varepsilon_\circ}\frac{q}{a} = -\frac{1}{4\pi\varepsilon_\circ}\frac{q^2}{a}$$

(iii) Work required to bring the charge +q to the corner R= q × potential at the point R due to charges at the point P and Q.

$$W_R = q \times \frac{1}{4\pi\varepsilon_\circ}\left(-\frac{q}{a}+\frac{q}{\sqrt{2}a}\right)$$

$$= \frac{1}{4\pi\varepsilon_\circ}\frac{q^2}{a}\left(-1+\frac{1}{\sqrt{2}}\right)$$

(iv) Work required to bring the fourth charge –q at the position S = q × potential at the point S due the all the three charges at the point P, Q and R

$$W_S = -q \times \frac{1}{4\pi\varepsilon_\circ}\left(\frac{q}{a}+\frac{q}{a}-\frac{q}{\sqrt{2}a}\right)$$

$$W_S = -\frac{1}{4\pi\varepsilon_\circ}\frac{q}{a}\left(2-\frac{1}{\sqrt{2}}\right)$$

(b) Work required to bring the charge q′ to the center of the square = q′ × potential at the center point O due to all the four charges in the four corners

The potential created by the two +*q* charges are canceled by the potential created by the -*q* charges which are located in the opposite corners. Therefore the net electric potential at the center O due to all the charges in the corners is zero.

Hence no work is required to bring any charge to the point O. Physically this implies that if any charge *q*′ when brought close to O, then it moves to the point O without any external force.

1.5.7 Electrostatic potential energy of a dipole in a uniform electric field

Consider a dipole placed in the uniform electric field $\vec{E}$ as shown in the Figure 1.31. A dipole experiences a torque when kept in an uniform electric field $\vec{E}$. This torque rotates the dipole to align it with the